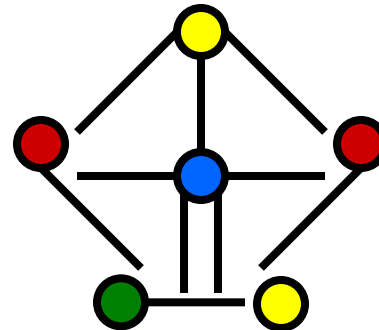
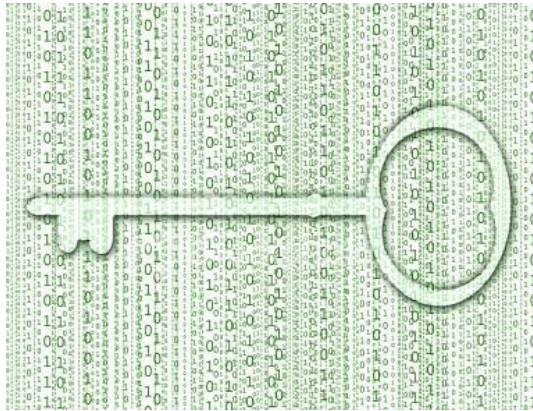
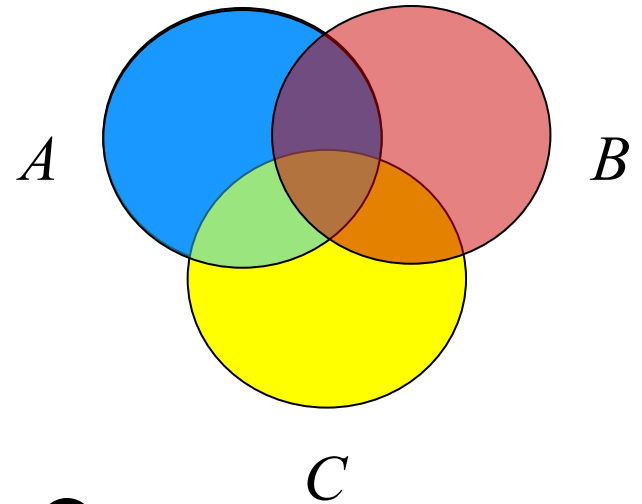


Introduction to Discrete Mathematics

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$



$$a = qb + r \implies \gcd(a, b) = \gcd(b, r)$$

Continued from Part A

	p	q	r	p or q	p or r	(p or q) and (p or r)	q and r	p or (q and r)
	0	0	0	0	0	0	0	0
	0	0	1	0	1	0	0	0
	0	1	0	1	0	0	0	0
	0	1	1	1	1	1	1	1
	1	0	0	1	1	1	0	1
	1	0	1	1	1	1	0	1
	1	1	0	1	1	1	0	1
	1	1	1	1	1	1	1	1

nent

DeMorgan

Distributive law

ties.

Theorem 1.1.1 Logical Equivalences

Given any statement variables p, q , and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

- | | | |
|--|---|---|
| 1. Commutative laws: | $p \wedge q \equiv q \wedge p$ | $p \vee q \equiv q \vee p$ |
| 2. Associative laws: | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ | $(p \vee q) \vee r \equiv p \vee (q \vee r)$ |
| 3. Distributive laws: | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |
| 4. Identity laws: | $p \wedge \mathbf{t} \equiv p$ | $p \vee \mathbf{c} \equiv p$ |
| 5. Negation laws: | $p \vee \sim p \equiv \mathbf{t}$ | $p \wedge \sim p \equiv \mathbf{c}$ |
| 6. Double negative law: | $\sim(\sim p) \equiv p$ | |
| 7. Idempotent laws: | $p \wedge p \equiv p$ | $p \vee p \equiv p$ |
| 8. Universal bound laws: | $p \vee \mathbf{t} \equiv \mathbf{t}$ | $p \wedge \mathbf{c} \equiv \mathbf{c}$ |
| 9. De Morgan's laws: | $\sim(p \wedge q) \equiv \sim p \vee \sim q$ | $\sim(p \vee q) \equiv \sim p \wedge \sim q$ |
| 10. Absorption laws: | $p \vee (p \wedge q) \equiv p$ | $p \wedge (p \vee q) \equiv p$ |
| 11. Negations of $\mathbf{t}$ and $\mathbf{c}$: | $\sim \mathbf{t} \equiv \mathbf{c}$ | $\sim \mathbf{c} \equiv \mathbf{t}$ |

Negation Exercise

- Jim is tall and thin
 - $\sim(\text{Jim is tall and thin})$
 - How to do this negation?
 - Jim is NOT tall and thin
 - Actually we should proceed as follows:
 - Original Statement: Jim is tall and Jim is thin
 - $\sim(\text{Jim is tall and Jim is thin})$
 - Jim is **Not** tall **or** Jim is **Not** thin
-

Tautology, Contradiction

A tautology is a statement that is always true.

$$p \vee \neg p$$

$$\underbrace{(p \wedge q)}_A \vee \underbrace{(\neg q \wedge p)}_B \vee \underbrace{(\neg p \wedge \neg q)}_C \vee \underbrace{(\neg p \wedge q)}_D$$



				A	B	C	D	
p	q	$\sim p$	$\sim q$	p and q	p and $\sim q$	$\sim p$ and $\sim q$	$\sim p$ and q	A or B or C or D
0	0	1	1	0	0	1	0	1
0	1	1	0	0	0	0	1	1
1	0	0	1	0	1	0	0	1
1	1	0	0	1	0	0	0	1

Tautology, Contradiction

A tautology is a statement that is always true.

$$p \vee \neg p$$

$$(p \wedge q) \vee (\neg q \wedge p) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge q)$$

A contradiction is a statement that is always false.

(negation of a tautology)

$$p \wedge \neg p$$

$$(p \vee q) \wedge (\neg q \vee p) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee q)$$

$$((p \wedge r) \vee (q \wedge r)) \wedge \neg(p \vee q) \\ \wedge r)$$



$$s = ((p \wedge r) \vee (q \wedge r)) \wedge (\neg(p \vee q))$$

p	q	r	$p \wedge r$	$q \wedge r$	$(p \wedge r) \vee (q \wedge r)$	$p \vee q$	$\neg(p \vee q)$	$\neg(p \vee q) \wedge r$	s
0	0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	0	1	1	0
0	1	0	0	0	0	1	0	0	0
0	1	1	0	1	1	1	0	0	0
1	0	0	0	0	0	1	0	0	0
1	0	1	1	0	1	1	0	0	0
1	1	0	0	0	0	1	0	0	0
1	1	1	1	1	1	1	0	0	0

Quick Summary

Key points to know.

1. Write a logical formula from a truth table.
2. Check logical equivalence of two logical formulas.
3. DeMorgan's rule and other simple logical rules (e.g. distributive).
4. Use simple logical rules to simplify a logical formula.